

Lesson:

Introduction



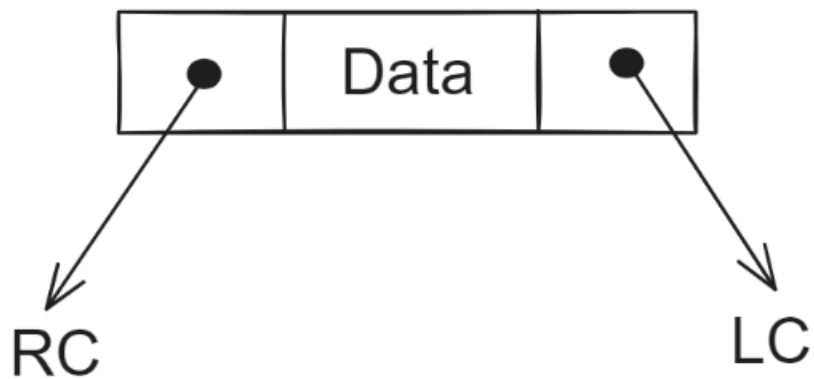
Topics

- Introduction to Trees
- Application of Trees
- Types of Trees

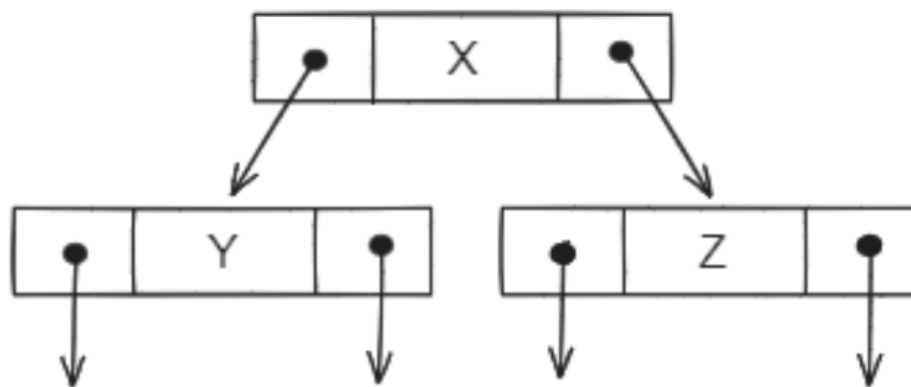
Introduction to Trees

A tree is a hierarchical data structure that consists of nodes connected by edges. It is a versatile data structure widely used in computer science for organizing and managing data in a hierarchical manner. The tree structure resembles an inverted tree with a root node at the top and branches connecting nodes, leading to the leaves at the bottom.

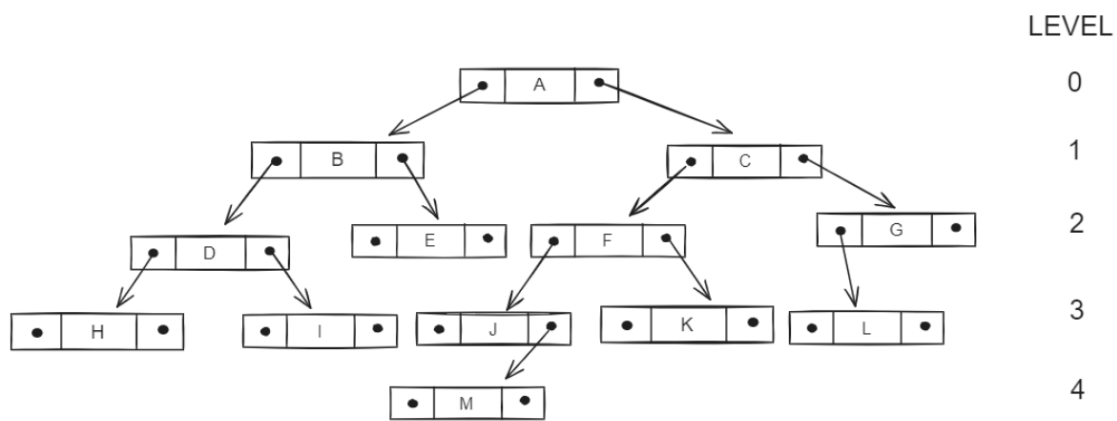
Basic Tree Terminology



(a) Structure of a node in a tree



(b) Parent, left child and right child of a node



(c) A simple tree with 13 nodes

Node

- This is the main component of any tree structure. The concept of the node is the same as that used in a linked list. A node of a tree stores the actual data and links to the other node. Figure (a) represents the structure of a node.

Parent and child

- A node's parent is the one that comes right before it, Here, X is the parent of Y and Z.
- If a node's parent is the one right before it, then all the nodes that come right after it are called its children. For example, in the figure below, Y and Z are the two children of X. The child which is on the left side is called the left child and that on the right side is called the right child.

Root:

- This is a specially designated node that has no parent. In Figure (c), A is the root node.

Link

- This is a pointer to a node in a tree. For example, as shown in Figure (a), LC and RC are two links of a node. Note that there may be more than two links of a node.

Leaf

- The node that is at the end and does not have any child is called the leaf node, in figure (c), H, I, K, L, and M are the leaf nodes, A leaf node is also alternatively termed the terminal node.

Level

- The level is the rank in the hierarchy. The root node has level 0. If a node is at level l , then its child is at level $l+1$ and the parent is at level $l-1$. This is true for all nodes except the root node, being at level zero, in Figure (c). The levels of various nodes are depicted.

Height

- The maximum number of nodes that is possible in a path starting from the root node to a leaf node is called the height of a tree. For example, in Figure (c), the longest path is A-C-F-J-M and hence the height of this tree is 5. It can be easily seen that $h = l_{\max} - 1$,

Where h is the height and $l_{\max}$ is the maximum level of the tree.

Degree

- The maximum number of children that is possible for a node is known as the degree of a node. For example, The degree of each node of the tree as shown in Figure (c) is 2.

Sibling

The nodes that have the same parent are called siblings, for example, in Figure (c), j and k are siblings.

Application of tree

File Systems:

Imagine your computer's file system as a tree. The root is the main directory, and each folder or file is a node branching off from it. This hierarchical structure allows for organized storage and retrieval of data.

Organizational Charts:

In a company, an organizational chart can be represented as a tree. The CEO is at the root, and each department and employee forms branches and leaves. This structure helps visualize reporting relationships and organizational hierarchy.

HTML Document Object Model (DOM):

Web pages are structured using HTML, and the DOM represents this structure as a tree. Each HTML element is a node, and their nesting creates the tree. This structure facilitates the manipulation and traversal of web page elements.

Expression Trees in Mathematics:

In mathematics, expressions like $(3 + 5) * 2$ can be represented as trees. The operators (+, *) are internal nodes, and the numbers are leaf nodes. This tree structure helps in evaluating and simplifying mathematical expressions.

Network Routing:

In computer networks, routing tables are often implemented using tree structures. Each node in the tree represents a routing decision, and the paths from the root to the leaves determine the routes for data packets.

Game Trees in Artificial Intelligence:

Game-playing algorithms, like those used in chess or tic-tac-toe, often employ trees to represent possible moves. Each node represents a game state, and branches represent possible moves, helping the AI determine the best strategy.

Family Trees:

Genealogical relationships in a family can be represented as a tree. Individuals are nodes, and connections between parents and children form branches. This structure simplifies the understanding of family relationships.

Decision Trees in Machine Learning:

Decision trees are used in machine learning for classification and regression tasks. They represent a series of decisions and their possible outcomes, helping to make predictions based on input features.

Huffman Coding for Compression:

Huffman coding, used in data compression algorithms, builds a binary tree to represent characters in a way that minimizes the total number of bits required for encoding, thus achieving compression.

XML/JSON Parsing:

Trees are commonly used to represent the hierarchical structure of XML and JSON data. Each tag or key-value pair becomes a node, forming a tree that simplifies the parsing and processing of structured data.

Types of Tree

1. Binary Tree

A binary tree is a tree data structure in which each node has at most two children, typically referred to as the left child and the right child.

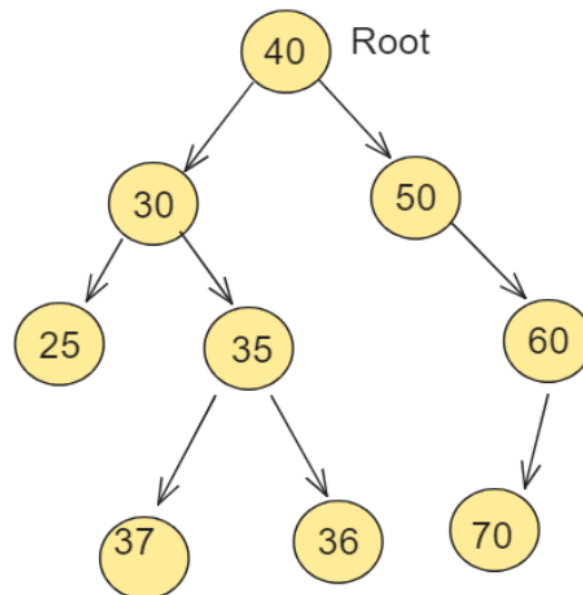
Types of binary tree

- Perfect binary tree
- Full binary tree
- Complete binary tree
- Degenerate binary tree
- Balanced binary tree

Note: we will discuss binary tree and their types in much detail in the next section

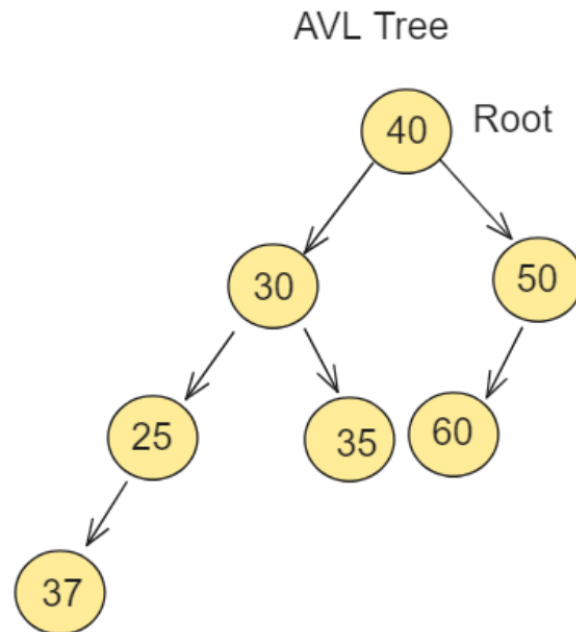
Binary Search Tree

Binary search Tree



- A binary tree data structure where each node has at most two child nodes, commonly referred to as the left child and the right child.
- The key property of a BST is that for each node, the values of nodes in the left subtree are less than or equal to the value of the node, and the values of nodes in the right subtree are greater than or equal to the value of the node.
- Binary Search Trees are often used for efficient search, insertion, and deletion

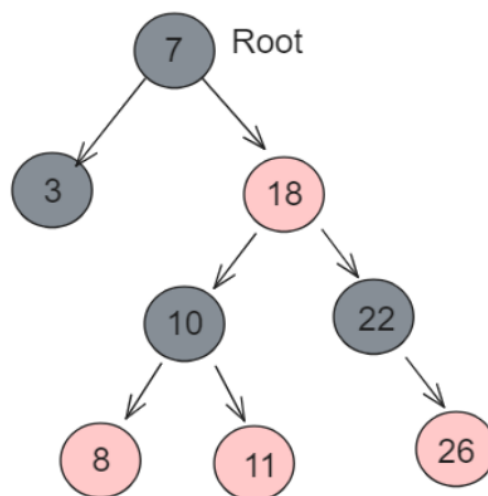
AVL Tree



- A self-balancing binary search tree (BST) named after its inventors Adelson-Velsky and Landis.
- In an AVL tree, the heights of the two child subtrees of any node differ by at most one.
- When a new node is inserted or a node is deleted, the tree automatically performs rotations to maintain the balance property, ensuring that the tree remains relatively balanced.
- The balancing property ensures that the height of the tree is logarithmic, leading to efficient search, insertion, and deletion operations with a time complexity of $O(\log n)$, where n is the number of nodes.

Red-Black Tree

Red Black Tree



- Another type of self-balancing binary search tree.
- Introduced by Rudolf Bayer in 1972 and later refined by others, including Leonidas J. Guibas and Robert Sedgwick.
- Red-Black Trees maintain balance using a set of color-coded rules for each node, ensuring that the tree remains balanced during insertions and deletions.
- The key properties include maintaining the black height of the tree (the number of black nodes on any path from the root to a leaf) and ensuring that certain rules about adjacent nodes' colors are not violated.
- Red-Black Trees provide efficient search, insertion, and deletion operations with a guaranteed logarithmic height.

B-Tree

- A self-balancing tree data structure that maintains sorted data and allows searches, insertions, and deletions in logarithmic time.
- B-Trees are commonly used in database systems and file systems for organizing and indexing data efficiently.
- Unlike binary trees, B-Trees are multiway trees, meaning each node can have more than two children.
- B-Trees maintain balance through a set of rules governing the number of keys and children each node can have, ensuring that the tree remains balanced even as data is inserted or deleted.
- The structure of B-Trees makes them well-suited for scenarios where data is stored on disk, as they minimize the number of disk I/O operations required for search and updates.

